

"Did you hear that?": Investigating localized spatial audio effects on enhancing immersive and affective user experiences in virtual reality

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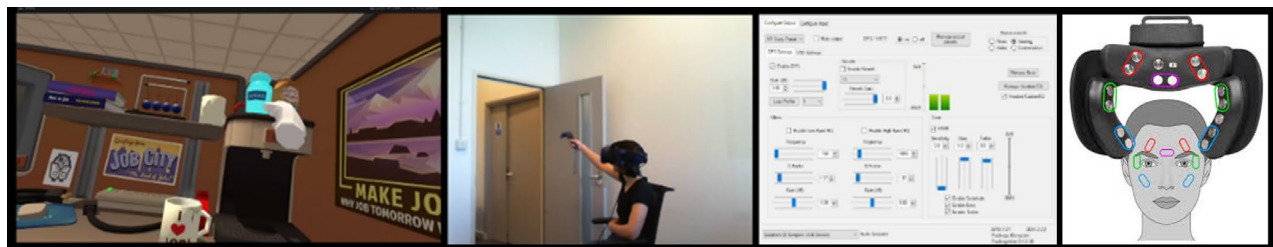


Figure 1. From left to right position: experiment view of the user in VR; user in the room; auditory enhancement software user surface, EmteqPRO headset; Abbreviations: EAC and E - enhanced audio condition; NC and N - normal audio condition.

ABSTRACT

This study examines the effects of software-based localised audio enhancement in VR on spatial audio, immersion, and affective responses. Sixty-eight participants took part in a VR game (Job Simulator) and reported improved sound quality, immersion, and localization under enhanced audio conditions. Physiological measures revealed significant effects on emotional valence, highlighting enhanced audio as a cost-effective way to improve auditory involvement and emotional engagement in VR without additional hardware.

Index terms: Audio systems, virtual reality, sensor systems human-computer interaction, electrophysiology, signal analysis.

1 INTRODUCTION

Virtual reality (VR) techniques have revolutionised the way users interact with simulated environments, providing high levels of immersion and the sensation of presence. Attaining high levels of immersion is crucial for creating a realistic and engaging experience, and one of the key components contributing to presence and immersion is spatial audio [1]. Properties of the auditory experience can enhance presence and realism in virtual

environments [2], especially when enabling users to hear localised sounds. As such, the applications of spatial audio effects in VR span various fields, including research, entertainment, training, digital-twin, and medical VR [3] leading to improved efficacy.

The creation of tools that allow for enhanced of audio quality at a software level could greatly benefit VR applications. Techniques such as binaural rendering and head-related transfer functions (HRTFs) can be employed to create a more immersive and realistic audio experience [4]. The study aim is to investigate the effects of a novel proprietary software-based spatial audio enhancement tool on the subjective auditory experience and related physiological measures of users in a standard VR experience. Affect was measured using the two-dimensional model of *valence* and *arousal* [5]. Physiological measures, continuously tracking changes within an VR experience, were heart rate measures (PPG sensor) linked to arousal and facial expression measures (facial electromyographic (EMG) sensors embedded within the headset) linked to pleasure (valence) [6] offering greater granularity in assessing user experience. It was predicted that enhanced compared to normal audio settings, firstly, increase audio quality, sound identification, sound involvement and sound localization (H1), and, secondly, to increase affect-based physiological responses related to valence and arousal (H2).

2 MATERIALS AND METHODS

Study description and VR content. Sixty-eight participants (mean age 28.65 ± 10.54 years) took part in the study. A two-minute physiological baseline was recorded, followed by a short EMG calibration to normalize the signals during data processing. After calibration, participants experienced the same VR experience twice, i.e. once with the EAC and once with NC condition, in a counterbalanced order. Here, they interacted with three virtual sound-emitting objects (a cafetiere, telephone, or copier) within the

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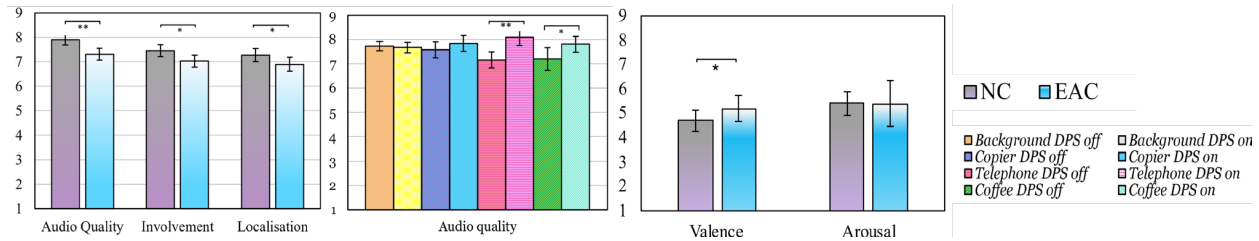


Figure 2. Average immersive auditory scores for the normal (NC) and enhanced audio (EAC) (left), individual event scores for audio quality (middle) and average valence and arousal detected levels. SE error bars. Significant differences indicated (*).

Job Simulator VR office worker version. The game was set in an indoor office environment with ambient office sounds and minimal distractions. It began with 30 seconds of passive observation before participants interacted with the objects in random order. After each interaction, participants completed a short auditory immersive experience questionnaire (items from [7]) on their auditory immersive experience. This study received ethical approval for the experimental protocol by the University of Geneva ethical commission (CUREG-2022-03-33).

Hardware & Software. VIVE Pro Eye VR headsets with integrated EmteqPRO masks were used for the VR application and the physiological data recording with the emteqPRO system [8] comprising a PPG and seven facial EMG sensors recorded at a sampling rate of 1000 Hz. Valence and arousal measures were extracted from the recorded data using emteq cloud analytics (described in [8]). The audio system used the built-in on-ear ‘Hi Res Audio Certified’ headphones.

The Audio Enhancement Tool. The Bongiovi APO audio enhancement tool installs directly onto the Windows audio driver of the HMD, enabling low-latency sound enhancement through the DPS and V3D audio processing solutions. These proprietary algorithms deliver a more natural and immersive listening experience tailored specifically for VR. The V3D solution uses custom Head-Related Transfer Function (HRTF) filters, calibrated in specialized acoustic labs, to provide precise sound localization based on human head anatomy. The V3D spatialization algorithm is combined with the Digital Power Station (DPS) algorithm, which enhances sound clarity and quality by accounting for perceived loudness rather than simple amplitude. This approach was specifically designed for optimizing auditory immersion in VR environments.

3 RESULTS & DISCUSSION

A 2x4 ANOVA (audio condition: EAC vs NC x 4 events) revealed significantly higher audio quality ratings for the EAC condition ($F(1, 63)=24.49, p<.001$; EAC Mean: 7.43 ± 0.35 vs NC Mean: 7.11 ± 0.36), with significant event-specific enhancements for the copier ($p=.045$) and telephone sounds ($p<.001$), but not for the cafetiere or background noise ($p's>.05$). Auditory presence ratings showed a significant main effect of audio condition for both involvement ($F(1, 63)=6.79, p=.011$) and localization ($F(1, 63)=5.95, p=.018$), with overall improvements in the EAC condition across all events. The event-specific enhancements suggest that audio quality improvements in the EAC condition were more pronounced for task-relevant sounds, while peripheral background noises remained unaffected. The cafetiere audio's consistent loudness across conditions indicates a ceiling effect, as it was perceived clearly in both NC and EAC conditions.

The enhanced audio condition (EAC) not only improved audio quality but also had a significant and notable effect on emotional

valence, making the experience more positive. Facial EMG measurements revealed that valence scores were significantly higher in the EAC condition (Mean = 5.2 ± 1.1) compared to the NC condition (Mean = 4.7 ± 0.86 ; Wilcoxon $t = 569.0, p = .0265$), highlighting how enhanced audio can positively influence affective responses beyond mere auditory perception. Interestingly, despite this clear impact on valence, arousal levels remained moderate and unaffected by audio enhancement (EAC Mean = 5.39 ± 1.90 , NC Mean = 5.42 ± 0.96 ; $p = .637$). These findings suggest that enhanced audio can influence emotional valence without necessarily increasing arousal, possibly due to the neutral and low-arousal nature of the office room environment. Future research will explore more stressful or emotionally engaging content to determine whether enhanced audio could similarly amplify arousal responses in higher-intensity scenarios.

4 CONCLUSION

This study highlights the significant impact of software-based audio enhancements in VR, improving immersion, sound localization, and emotional valence, particularly during focused events. While valence responses became more positive, arousal levels remained unaffected, suggesting that the emotional impact of enhanced audio may depend on content. Future research should integrate automatic event annotation tools and explore individual differences in audio sensitivity to optimize personalization. By advancing these methods, VR applications can further enhance user engagement, and overall immersive experiences.

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